

Probability Review

The toolbox beneath the course — and where each tool was used

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A handful of facts about *expectation*, *Bayes* and the *Gaussian*, carrying an entire course

— ORIENTATION

Orientation

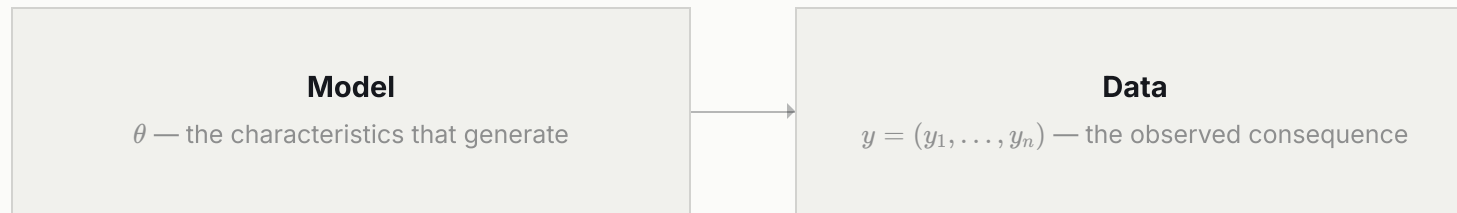
Not a prerequisite to be read first. A ledger, to be read afterwards — with a pointer to the slide where each fact did the work.

Why an appendix — the few facts everything rested on

Every method this term leaned on a handful of probability facts. Eight rows, and [a pointer to where each one carried the weight](#).

TOOL	WHERE IT CARRIED THE WEIGHT
expectation, and linearity	every return and value (Ch 7–11); expected improvement (Ch 4)
law of total expectation	<i>the prior mean averages every posterior mean</i> (Ch 2); Bellman (Ch 7)
law of total variance	<i>the posterior is on average tighter</i> (Ch 2); baselines (Ch 10)
Bayes' rule	the whole of belief updating (Ch 2–4); CbAS's target (Ch 6)
the multivariate Gaussian	the weight posterior (Ch 2); GP conditioning (Ch 4); LQR noise (Ch 9)
Monte Carlo estimation	every sampled Bellman backup (Ch 8); REINFORCE (Ch 10)
importance sampling	the CbAS ladder (Ch 6); off-policy evaluation (Ch 12)
KL divergence, entropy	the VAE's regulariser (Ch 6); TRPO's trust region (Ch 10)

Model to data, and the arrow inference runs backward



Statistics infers the causes that generated the observed data. The arrow runs left to right in the world, and right to left in inference.

Chapter 2 opened on exactly this picture and never left it: a coin's bias generates a sequence of tosses; the tosses are what we see. Chapter 3 gave θ a graph, Chapter 4 made it an entire function, and Chapters 7–12 made it a transition kernel. The arrow never changed direction — only the size of the thing at its tail.

Each tool below is one line of algebra with an outsized payoff. **The pointer beside it is the point of the appendix.**

— EXPECTATION

Expectation and its laws

Almost every objective in this course is an expectation. Two laws about *conditioning* do most of the work.

Mean, variance, and the mean of a function

For a random variable U with density p , and any $g : \mathbb{R} \rightarrow \mathbb{R}$:

$$\mathbb{E}[U] = \int u p(u) du, \quad \mathbb{E}[g(U)] = \int g(u) p(u) du, \quad \text{var}(U) = \mathbb{E}[(U - \mathbb{E}[U])^2] = \mathbb{E}[U^2] - \mathbb{E}[U]^2$$

Linearity is the property used most often, and the one most often used without noticing:

$$\mathbb{E}[aU + bV] = a \mathbb{E}[U] + b \mathbb{E}[V] \quad \text{regardless of any dependence between } U \text{ and } V.$$

It is why a return can be split term by term, why a baseline can be subtracted from a policy gradient without touching its mean, and why the covariance of AY is $A\Sigma A^\top$. Every objective the course optimised is an expectation: the return $\mathbb{E}_\pi[\sum_t \gamma^t r_t]$ (Ch 7–11), expected improvement $\mathbb{E}[\max(0, f - f^{\max})]$ (Ch 4), the ELBO $\mathbb{E}_q[\log p(x | z)]$ (Ch 6), the policy objective $\mathbb{E}_\tau[R(\tau)]$ (Ch 10). **Manipulating expectations is the technical core of the course.**

The two laws that recur

LAW OF TOTAL EXPECTATION

average the conditional averages

$$\mathbb{E}[U] = \mathbb{E}_V[\mathbb{E}[U | V]]$$

LAW OF TOTAL VARIANCE

uncertainty splits in two, and the split is exact

$$\text{var}(U) = \underbrace{\mathbb{E}_V[\text{var}(U | V)]}_{\text{within — noise left inside each condition}} + \underbrace{\text{var}(\mathbb{E}[U | V])}_{\text{between — spread across conditions}}$$

Both say the same thing from two heights: knowing V does not change your average, and it does reduce your spread — by **exactly the amount your average moves around** as V varies. Everything on the next two slides is those two sentences, cashed.

Total expectation, read in three chapters

CH 2 — THE PRIOR MEAN

$$\mathbb{E}[\theta] = \mathbb{E}[\mathbb{E}[\theta \mid y]]$$

Before seeing data, your belief is already the average of every belief you might end up with. A posterior that moved in one direction *for every possible dataset* would be an incoherent prior.

CH 7 — THE BELLMAN EQUATION

$$V^\pi(s) = \mathbb{E}[r_t + \gamma V^\pi(s_{t+1}) \mid s]$$

Lecture 7 peeled one step off the return and let the Markov property close the loop. **That peel is the tower rule** — condition on the next state, average the conditional averages.

CH 8 — THE SAMPLED BACKUP

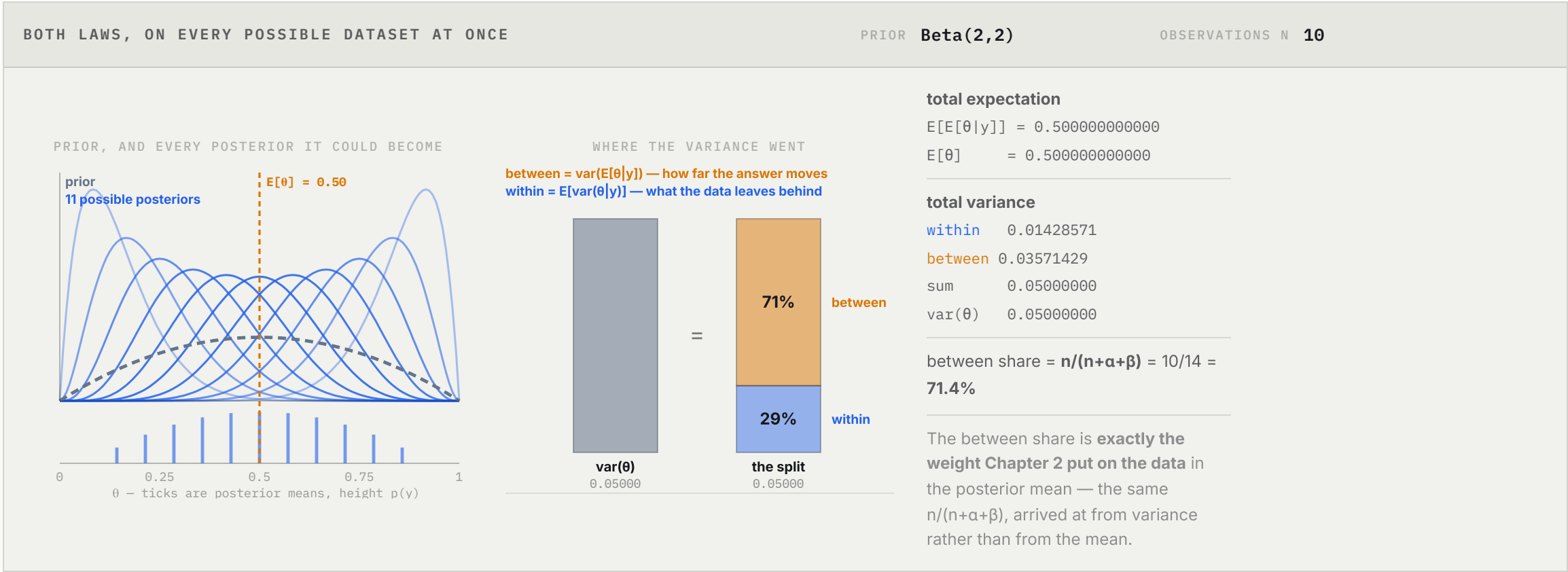
$$\mathbb{E}_{s'}[\cdot] \longrightarrow \text{one transition } (s, a, r, s')$$

Lecture 8's translation table swaps the *outer* expectation for a single draw and calls the law of large numbers to justify it. The equation is unchanged; only who evaluates it.

↪ used in Ch 2 (the prior–posterior identity), Ch 7 (every Bellman equation), Ch 8 and 10 (every sampled backup).

Total variance, on every dataset at once

the identity computed rather than asserted — every dataset enumerated, nothing sampled



The three places that split mattered

CH 2 – THE POSTERIOR TIGHTENS

$\text{var}(\theta \mid y)$ is smaller than $\text{var}(\theta)$ by exactly the spread of posterior means. Chapter 2 stated it as **data cannot, on average, make you less certain** — which is this identity read as an inequality.

CH 2 AND 4 – THE PREDICTIVE SPLIT

The predictive variance is $\sigma_Y^2 + \tau_1^2$: measurement noise you can never remove, plus parameter uncertainty you can. Lecture 0's aleatoric/epistemic pair, arriving as the two terms of one identity.

CH 10 – THE BASELINE

Subtracting $\mathbb{E}[U \mid V]$ deletes the between part and leaves the mean untouched. That is why a baseline reduces the variance of a policy gradient at **zero bias** — Chapter 10 measures the effect at nearly twentyfold — and why the advantage exists at all.

One identity, three readings: *how much did I learn*, *what can I never learn*, and *how do I stop the estimator from shaking*. Whenever a later chapter claims a quantity is "smaller on average" or "unbiased but less noisy", this is the line being invoked.

— BAYES

Bayes and the two views

One line of algebra that turns a belief and a measurement into a new belief — and two incompatible ways of saying what "uncertain" means.

Bayes' rule — the whole of inference on one line

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) p(\theta)}{p(\text{data})} \propto \underbrace{p(\text{data} \mid \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

WHERE IT WAS THE WHOLE METHOD

- **Ch 2** — prior to posterior, conjugacy, and MAP as MLE plus a prior term. Regularisation turned out to *be* a prior.
- **Ch 3** — the same rule with many unknowns at once: marginalise the hidden variables, then normalise. The graph writes the joint as a product; *summing it back down* is where the cost lives.
- **Ch 4** — a prior over *functions*; the GP posterior is Bayes' rule with θ replaced by f .

WHERE IT WAS HIDING

- **Ch 6** — CbAS writes its target as $p(x \mid S) \propto P(S \mid x) p(x \mid \theta^{(0)})$: **the generative model is the prior and the oracle is the likelihood.**
- **Ch 5** — an ensemble's disagreement is a crude posterior width, used as an out-of-distribution alarm.

Belief in = prior; belief out = posterior; **the data does the turning.**

Frequentist and Bayesian — two readings of the same data

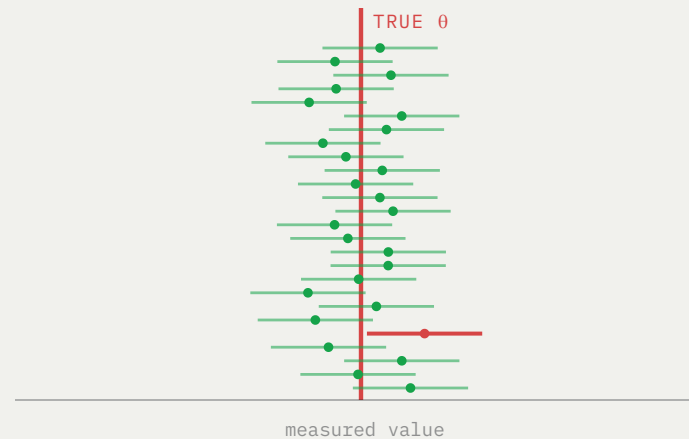
	FREQUENTIST	BAYESIAN
what is random	θ fixed, the data random	the data fixed, θ random
what varies	variability <i>in the data</i> given θ	variability <i>in θ</i> given the data
the interval	confidence: $\theta \in \text{CI}$, or not	credible: $\Pr(\theta \in \text{CR}) = 0.95$
the sentence	95% of repeated intervals cover θ	95% chance θ lies in <i>this</i> region

The distinction is not pedantry. It is the MLE-versus-MAP, point-versus-distribution choice that Chapter 2 was built on, and it is why a Gaussian process reports a credible **band** rather than a confidence *statement*.

Both say 95%, and they are different claims

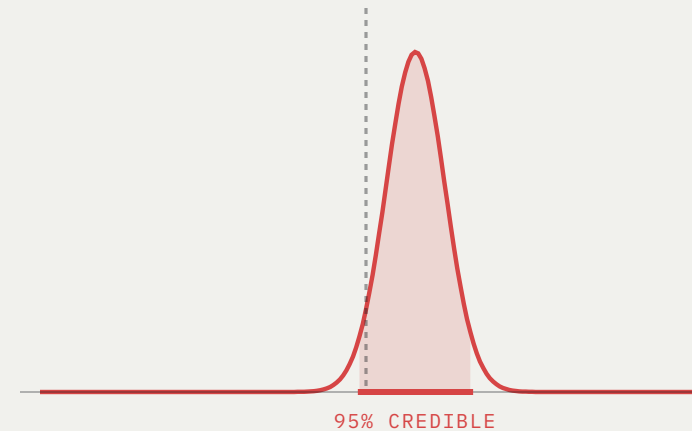
BOTH SAY 95% — AND THEY ARE DIFFERENT CLAIMS

FREQUENTIST — VARIABILITY IN THE DATA



25 of 26 intervals cover the true θ .
The 95% is a property of **the procedure**, not of any one interval.

BAYESIAN — VARIABILITY IN θ



From **this one dataset**: $\Pr(\theta \in \text{CR}) = 0.95$.
The 95% is a property of θ , given what was seen.

Left, the frequentist picture: θ is a fixed vertical line and the *intervals* are what move — run the experiment 26 times and about 95% of them cover it. Right, the Bayesian picture: one dataset, and the *parameter* is what is spread. Chapter 2 ran this widget for the same reason.

The other two views — aleatoric and epistemic

ALEATORIC — IRREDUCIBLE

The world is genuinely stochastic. Repeating the experiment gives a different answer, and no amount of data removes it. All you can do is *characterise* it, as a distribution.

Measurement noise σ_Y^2 (Ch 2), observation noise σ_ϵ^2 on the GP diagonal (Ch 4), the transition kernel's own randomness (Ch 7).

EPISTEMIC — REDUCIBLE

We simply do not know the model: its parameters, its state, its dynamics. **Data shrinks it, and the whole course is a campaign against it.**

Parameter uncertainty τ_1^2 (Ch 2), the GP band away from data (Ch 4), ensemble disagreement (Ch 5), and PETS' ensemble spread far from data, whose *per-member* variance is the aleatoric half (Ch 11).

Lecture 0 named the pair on the slide *Two kinds of uncertainty — name them once*; Chapter 2's Act 4 ended by writing them as the two terms of $\sigma_Y^2 + \tau_1^2$. That is the law of total variance again — the *within* part is aleatoric, the *between* part epistemic — which is why this appendix keeps circling the same identity.

— GAUSSIAN

The multivariate Gaussian — the workhorse

The single most-used distribution in the course. Four properties, and each one buys a chapter.

The multivariate normal — definition

A random vector $Y = (Y_1, \dots, Y_k)$ is jointly Gaussian if

$$p(y) = \mathcal{N}(y \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \exp\left(-\frac{1}{2}(y - \mu)^\top \Sigma^{-1}(y - \mu)\right)$$

with $\mathbb{E}[Y] = \mu$ and $\text{var}(Y) = \Sigma \succeq 0$, $\Sigma_{ij} = \text{cov}(Y_i, Y_j)$.

Four properties make it the backbone of [Gaussian-process regression](#) — and of half the closed forms in this course.

Each is a one-line fact. Together they are why so much of the course can carry uncertainty *analytically* rather than by sampling: a Gaussian goes in, a Gaussian comes out, and the parameters update by matrix algebra rather than by an integral you cannot do.

Properties 1 and 2 — independence and linearity

PROPERTY 1 — FOR GAUSSIANS, UNCORRELATED MEANS INDEPENDENT

Setting $\Sigma_{ij} = 0$ for $i \neq j$ makes Σ and Σ^{-1} diagonal, so the joint density factors:

$$\Sigma \text{ diagonal} \implies p(y) = \prod_i \mathcal{N}(y_i \mid \mu_i, \sigma_{ii})$$

For general random variables this is false — take

$$X \sim \mathcal{N}(0, 1)$$

and

PROPERTY 2 — A LINEAR MAP OF A GAUSSIAN IS GAUSSIAN

$$Z = AY \sim \mathcal{N}(A\mu, A\Sigma A^\top)$$

Both moments are just linearity:

$$\mathbb{E}[AY] = A\mu$$

and

$$\text{cov}(AY) = A\Sigma A^\top$$

.

↪ the reparameterisation trick $z = \mu_\phi + \epsilon\sigma_\phi$ (Ch 6), and linear-Gaussian dynamics $x_{t+1} = Ax_t + Bu_t + w_t$ (Ch 9).

Properties 3 and 4 — marginals and conditionals stay Gaussian

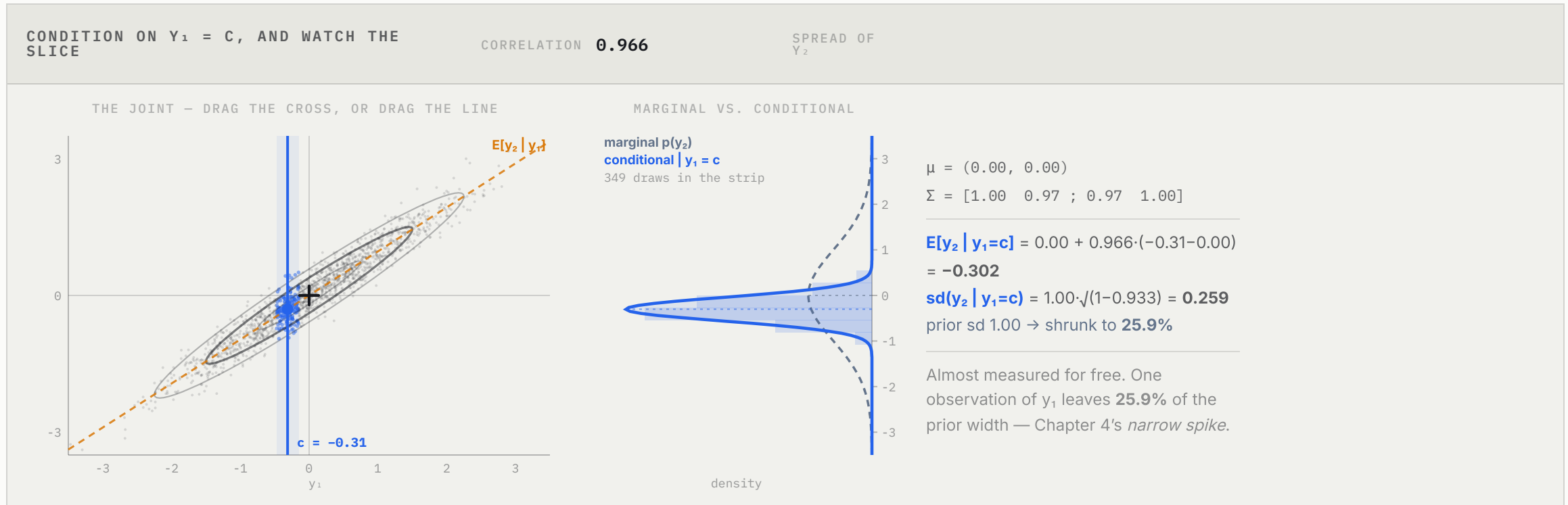
Property 3 — marginals are Gaussian. Any sub-vector of a Gaussian is Gaussian; read off the matching blocks of μ and Σ and ignore the rest. (It is Property 2 with a selector matrix.)

Property 4 — conditionals are Gaussian. The one that makes GP regression work. For

$$Z = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right) \implies Y_2 \mid Y_1 = y \sim \mathcal{N}\left(\underbrace{\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(y - \mu_1)}_{\text{posterior mean}}, \underbrace{\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}_{\text{posterior covariance}}\right)$$

Read the two braces. The mean moves by [how far the observation fell from its own mean, geared by the correlation](#); the covariance drops by a term that does not depend on the observed *value* at all — which is why a GP's error bars can be planned before any data arrives, and why Chapter 4 could choose where to sample next.

Conditioning, in your hands



Drag the mean, reshape Σ with the correlation and scale controls, then drag the conditioning line $y_1 = c$. The blue slice is the exact conditional: its mean slides along the regression line $\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(c - \mu_1)$ and its width shrinks to $\sigma_2 \sqrt{1 - \rho^2}$ — **a factor that depends on the correlation and on nothing else**. The two presets are Chapter 4's own numbers.

Where Property 4 did the work

CH 4 — THE GP POSTERIOR

Take $Y_1 = \mathbf{y}_{1:n}$ and $Y_2 = f(x)$, so $\Sigma_{11} = \mathbf{K} + \sigma_\epsilon^2 \mathbf{I}$, $\Sigma_{21} = \mathbf{k}^\top$, $\Sigma_{22} = k(x, x)$:

$$\mu(x \mid \mathcal{D}) = \mathbf{k}^\top (\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}$$

A GP posterior is Gaussian conditioning, and nothing else.

CH 2 — THE WEIGHT POSTERIOR

With $y = \mathbf{X}w + \epsilon$ and both w and ϵ Gaussian, w and y are *jointly* Gaussian — so $p(w \mid \mathbf{X}, y)$ is Property 4 again.

It lands on $\mathcal{N}(\mu_w, \Sigma_w)$ with μ_w the ridge solution, $\lambda = \sigma^2 / \alpha^2$.

CH 9 — LQR WITH NOISE

The value is quadratic, $V(x) = x^\top P x$. Put Gaussian noise back into $x_{t+1} = Ax_t + Bu_t + w_t$ and

$$\mathbb{E}[x^\top P x] = \mu^\top P \mu + \text{tr}(P \Sigma), \quad x \sim \mathcal{N}(\mu, \Sigma)$$

The noise adds a *constant*. The gain K is unchanged — certainty equivalence, in one line.

Chapter 4 quantified this at two correlations from its own kernel. At $\text{corr}(f_1, f_2) = 0.966$ one observation shrinks the neighbour's standard deviation to $\sqrt{1 - 0.966^2} = 0.259$ of the prior — *a narrow spike*. Four points apart, at $\text{corr} = 0.573$, it shrinks only to 0.820 — *barely narrower than the prior*, a value you still have to buy. Both are $\sqrt{1 - \rho^2}$ and nothing more.

— SAMPLING

Sampling and divergences

What to do when the integral has no closed form — and how to say that one distribution is far from another.

Monte Carlo — when the integral is intractable

When an expectation has no closed form, **estimate it by sampling**:

$$\mathbb{E}_{x \sim p}[g(x)] \approx \hat{\mu}_N = \frac{1}{N} \sum_{i=1}^N g(x_i), \quad x_i \sim p$$

unbiased by construction, with standard error $\sigma_g/\sqrt{N}$ by the central limit theorem — where $\sigma_g^2 = \text{var}_p(g)$.

This is the quiet foundation of *model-free* reinforcement learning. When the expectation over unknown dynamics cannot be computed, average over sampled trajectories instead: Monte-Carlo returns and TD targets (Ch 8), REINFORCE and advantage estimates (Ch 10), and the ELBO's reconstruction term, which the reparameterisation trick turns into a one-sample average (Ch 6). ↪ *Ch 8 and 10 — the entire data-driven half of the course.*

For an i.i.d. average the rate $1/\sqrt{N}$ is fixed by the central limit theorem. **The constant σ_g is entirely yours to choose.**

Importance sampling — sample here, answer about there

$$\mathbb{E}_{x \sim p}[g(x)] = \mathbb{E}_{x \sim q} \left[\underbrace{\frac{p(x)}{q(x)}}_{\text{weight } w(x)} g(x) \right] \quad \text{for any } q \text{ with } q > 0 \text{ wherever } pg \neq 0$$

WHERE IT WORKED — CH 6

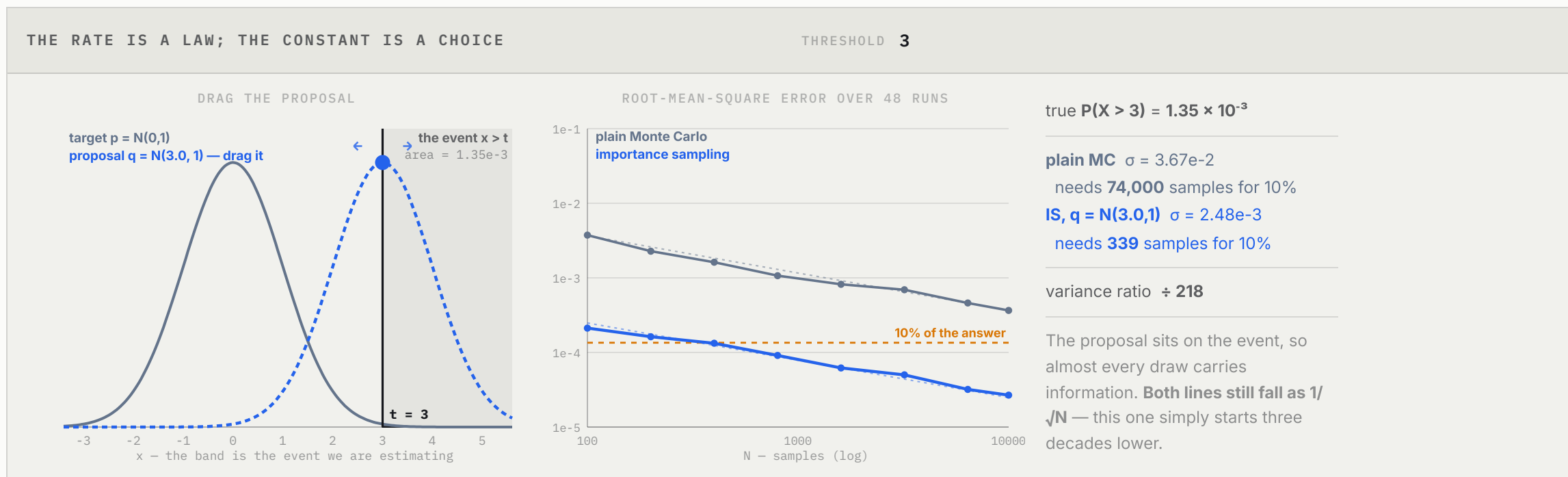
CbAS cannot draw from $p(x \mid S)$ directly, so it draws from the previous iterate and reweights. The ladder exists precisely to keep q close to the target, **so the weights never get the chance to explode**.

WHERE IT IS THE WHOLE PROBLEM — CH 12

Off-policy evaluation scores a policy you may not deploy, by reweighting a dataset another policy collected. The weight is a *product over the horizon*, so its variance grows with T — which is why the chapter reaches for doubly-robust estimators.

Unbiased for *any* legal q . The catch is entirely in the variance: $\text{var}_q(wg)$ can be far smaller than $\text{var}_p(g)$, far larger, or infinite — and unbiasedness will not warn you which.

The rate is fixed; the proposal sets the constant



Estimating $\Pr(X > 3) = 1.35 \times 10^{-3}$ for $X \sim \mathcal{N}(0,1)$. Both clouds of points fall as $1/\sqrt{N}$ and sit on their own exact $\sigma/\sqrt{N}$ line: the rate is not negotiable. What moves is the constant — a proposal aimed at the event needs **339 samples for 10% accuracy**, plain sampling needs **74,000**, and one aimed the wrong way needs **3.4 million** and returns a confident zero until it does not.

KL divergence and entropy — comparing distributions

$$\text{KL}(p \parallel q) = \mathbb{E}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right] \geq 0, \quad = 0 \iff p = q$$

Not symmetric, and not a metric — but the natural *information cost* of using q where p is true. For Gaussians it is closed form, which is why it can sit inside a loss: $\text{KL}(\mathcal{N}(0, 1) \parallel \mathcal{N}(0, 2)) = 0.318$ while the reverse is 0.807.

- **Ch 6, the VAE.** The ELBO's KL term pulls the encoder onto the prior, which is what makes the latent space **samplable**: without it q_ϕ is free to encode anywhere and new draws decode to nothing.
- **Ch 6, generative training.** Minimising KL to the data distribution *is* maximum likelihood; minimising JS instead is the GAN. Same shape, different divergence, different failure mode.
- **Ch 10, trust regions.** TRPO maximises an importance-weighted advantage subject to $\mathbb{E}[\text{KL}(\pi_{\text{old}} \parallel \pi_\theta)] \leq \delta$ — the largest step that keeps the new policy *close in behaviour*, not close in parameters. PPO recovers the effect with a clip.

Entropy $H(p) = -\mathbb{E}_p[\log p(x)]$ measures spread rather than distance. Maximising it keeps a policy exploratory — the seed of maximum-entropy RL — and it is the term that turns a hard argmax into a soft, probabilistic best response.

— CLOSING

Closing

Everything above, on one page.

The toolbox map — one glance

PROBABILITY TOOL	LECTURES THAT DEPEND ON IT
expectation, linearity	all of Part IV — returns and values
law of total expectation	the Bellman equation (Ch 7–11)
law of total variance	the predictive split (Ch 2, 4); baselines and advantage (Ch 10)
Bayes' rule	Ch 2, 3, 4; CbAS's target (Ch 6)
Gaussian, Property 4	Ch 4 — the GP posterior <i>is</i> this formula
Gaussian, linear map	Ch 6 (reparameterisation), Ch 9 (LQR noise)
Monte Carlo estimation	Ch 8, 10 — model-free RL
importance sampling	Ch 6 (CbAS), Ch 12 (off-policy evaluation)
KL, entropy	Ch 6 (VAE), Ch 10 (TRPO)

Two objects, and the rest is bookkeeping

The whole course manipulates an **expectation** — what we believe will happen on average — and a **Gaussian** — the one uncertainty we can carry in closed form.

Everything else is how to **update** them (Bayes, conjugacy, conditioning), how to **estimate** them when the integral will not close (Monte Carlo, importance sampling), or how to **bound the gap** between two of them (KL, trust regions).

Keep this appendix beside the main lectures. Whenever a derivation moves quickly through a probability step — a swapped expectation, a conditional that stays Gaussian, a sample average standing in for an integral — the justification is one of the ten rows above, and the pointer beside it says where to look.

An expectation, and a Gaussian.

Update them, estimate them, or bound the gap between them — that is the course.